

Digital Services Act

Transparency Report for Outlook

February 2025

Introduction

This report covers Outlook. Outlook is a productivity tool used by customers to send and receive emails, schedule meetings, manage Groups, among other productivity tasks.

This DSA Transparency Report is responsive to the obligations under DSA Article 15(1), Article 24(1), and Article 42(2)-(3). This Report provides information regarding the following topics as they pertain to the European Union:

- Content Moderation at Microsoft's Initiative
- Content Moderation & Automated Systems
- Content Moderation following a User Report
- Content Moderation Appeals
- Government Requests

Content Moderation at Microsoft's Initiative

Microsoft provides the information below in response to DSA Article 15(1)(c). This section reports data regarding content moderation Outlook engaged in on its own initiative, absent a user report.

At its own initiative, Microsoft performs hash matching that leverages the PhotoDNA and MD5 technologies to prevent the dissemination of Child Sexual Exploitation and Abuse (CSEA) using Outlook. In Outlook, this hash matching is currently done for attachments in outbound emails in the North American (NAM) geo. If the hash set identifies a match, the email is blocked from being sent. Review by trained human reviewers is used to confirm the accuracy of the technology and ensure that the appropriate enforcement actions are taken in response to matches. If content is deemed against CSEAI policy, the email sender's account is suspended and reported to the proper authorities as legally required.

The automated detection methods leveraged by Outlook are hash matching using PhotoDNA and MD5 hashes for the purpose of detecting CSEA and TVEC.

PhotoDNA:

PhotoDNA is a perceptual hashing technology, which is used to identify visually identical and near identical images and video, which Microsoft uses to detect known CSEA and Terrorism content. PhotoDNA creates a unique, not reversible, digital signature (known as a “hash”) of an image or selected frames in a video and compares it against signatures (“hashes”) of previously identified CSEA and Terrorism Content images or video. PhotoDNA is not facial recognition software and cannot be used to identify a person or object in an image. It does not use AI or machine learning.

Since its development in 2009, it has been used by hundreds of organizations globally and has assisted in the detection, disruption, and reporting of millions of child exploitation images. You can find more information about PhotoDNA here:
<https://www.microsoft.com/en-us/photodna>.

Although PhotoDNA was developed for the specific purpose of combatting the distribution of CSEA through online platforms, the technology is agnostic as to content-type and has been used to detect other visually identical and near identical images and videos. With limited exceptions, Microsoft has, by license, limited its use to CSEA, Terrorism, and NCII content.

For CSEA, PhotoDNA is used to detect content that has previously been determined to sexualize a child based on a ratings criteria widely adopted across industry. It draws from an internal database of Microsoft-created hashes for CSEA content, as well as from the following trusted third-party organizations and their hash databases: the Internet Watch Foundation, the US National Center for Missing and Exploited Children (NCMEC), which maintains its own databases and manages a database that enables industry hash sharing), Cybertip.CA, the Royal Canadian Mounted Police (“RCMP”), and Meta (which manages ThreatExchange, another industry hash sharing database).

For Terrorism Content, PhotoDNA is used to detect content that has previously been determined to be terrorism and violent extremism content consistent with Microsoft policies and which has been hashed for inclusion in an internal database of Microsoft-created hashes, as well as from the Global Internet Forum to Counter Terrorism (“GIFCT”).

MD5:

Message-Digest Algorithm (“MD5”) is a broadly used cryptographic hashing technology, which is provided as part of standard programming libraries and is used to determine if two items of content are identical. Microsoft currently deploys MD5 in conjunction with PhotoDNA to identify CSAM and Terrorism Content. Microsoft neither developed nor acquired ownership of MD5. More information about MD5 is available on [Wikipedia](#). More information on how Microsoft leverages MD5 can be found here:

<https://www.microsoft.com/en-us/DigitalSafety/moderation-and-enforcement/content-detection>.

Content Moderation & Automated Systems

Microsoft provides the information below in response to DSA Articles 15(1)(e) and 42(2)(c).

There are two primary contributors to the effectiveness of hash matching systems that leverage PhotoDNA or MD5: the correctness of the hashing algorithm itself (the technology) and the management of the set of hashes against which content is matched (the content). The technology itself is content agnostic.

The correctness of the hashing algorithm itself is ensured by validating that any updates do not alter its results on a known set of inputs (i.e., we confirm that the hash calculations return the same result, which is necessary to achieve 100% recall).

Hash matching leverages databases of reference hashes of known CSEA and Terrorism images or videos when assessing potentially violative content to find copies of the same content.

The following information relates to how the performance of the content (i.e. reference hashes) is tested and monitored. This is separate and distinct from the performance of the technology and is impacted by match distances used and the quality and consistency of human reviews both by Microsoft and by third parties that generate hashes that Microsoft leverages.

Reference hashes are updated and maintained in the following ways:

- Human review improves the accuracy of the underlying hashes by considering what the reviewer determined.

- In addition, in the case of PhotoDNA, hashes that are very similar are grouped to ensure they provide the same result.

As hash sharing entities provide new hashes, they are merged with the database to ensure consistent results.

Microsoft has no specific acceptance threshold for the rate at which matched content is determined to be CSEA beyond our goal to improve the rate over time. Microsoft targets a rate of less than 5% for matched content that is determined to be benign, because historically, it has been able to achieve that rate without losing recall. Given the nature of hash matching and the inherent differences of both policies and training between Microsoft and our industry peers, we cannot expect 100% precision. As a result, our thresholds are informed by historical performance and we carefully weigh any improvements in precision in either rate against the potential for loss of recall.

Outlook sends the content to be scanned and its associated metadata to the central service.

The central service orchestrates any and all human reviews that are required by policy and, if the content is confirmed to be violative, the centralized service will communicate the need to take any associated content-level actions (e.g. blocking sharing or deleting the content) back to the in-scope service.

The central service then performs any additional required processing including notification of the action to the content owner, account suspension (if warranted), reporting any CSEA discovered to NCMEC, and persisting the evidence to enable adjudicating appeals and responding to any resulting law enforcement data requests.

The precise purpose of these automated means is to prevent the dissemination of known CSEA on Outlook.

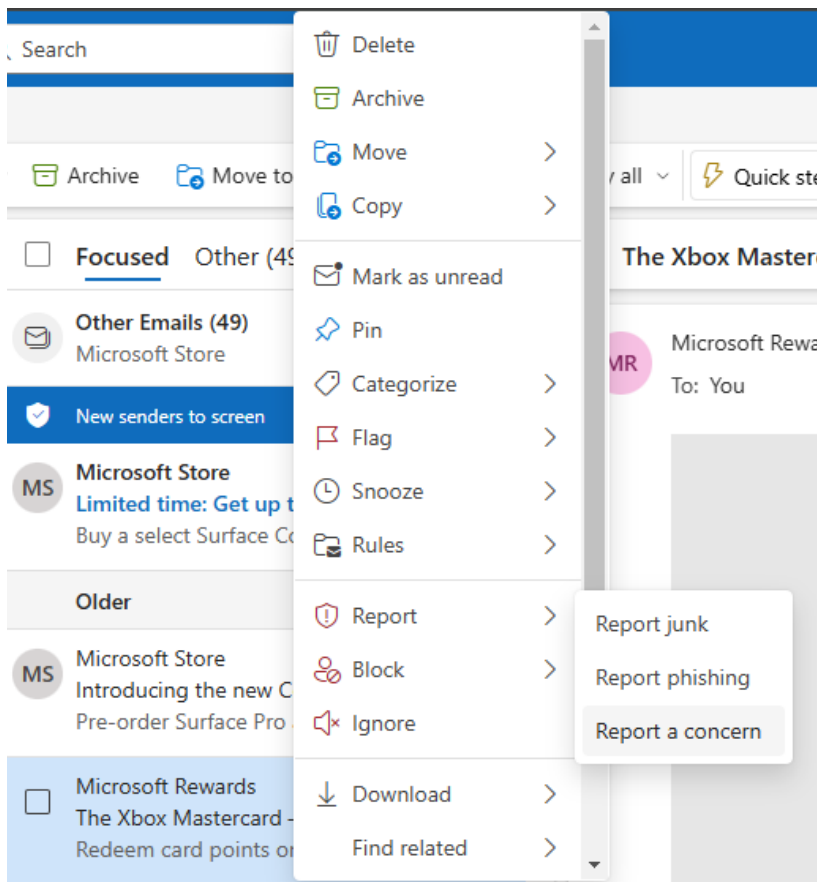
Microsoft leverages a broader human review process in conjunction with hash-based detection. A matched image is directed to a human moderation queue (depending on the harm type of the matching reference hash) where one or more moderators may independently review the content for quality control purposes. In order for a hash signature to be added to Microsoft's internal database of verified hashes, the associated image or video frame must have been reviewed by at least two human reviewers.

	Outlook.com
Number of measures not taken by automated means	66

Content Moderation following a User Report

Microsoft provides the information below in response to DSA Articles 15(1)(b)-(c) and 41(2)(a)-(b).

In Outlook, a consumer user can report a concern for an email by right clicking on the email and selecting Report > Report a concern:



This launches the report a concern dialog:

How does this content violate the Microsoft Services Agreement?

For reference, see the [Code of Conduct](#).

☐ Sexually inappropriate
☐ Child sexual exploitation or abuse
☐ Terrorist or violent extremist content
☐ Imminent harm to persons or property
☐ Hate speech
☐ Non-consensual intimate imagery
☐ Copyright infringement

Comments

Please provide as much detail as possible regarding the abuse or offensive behavior you are reporting. Please do not provide any personal or sensitive information.

☐ Keep my report anonymous

I agree that by clicking Report, I am submitting a copy of the offending email (including metadata) which may contain personal information and will be handled in accordance with the terms of the [Microsoft Privacy Statement](#).

Policy Violation	Number of notices received	Number of specific items of information included in the total number of notices	Median time to take action	Number of actions taken on the basis of the terms and conditions of the service
Illegal or Harmful Speech	268	268		
IP Infringements	62	62	86.3	
Nonconsensual Behaviour	3	1	265.9	53
Pornography or Sexualized Content	1,388	1,388	5.8	12
Protection of Minors	22	22		
Risk for public security	59	59		
Scams and Fraud	10	10	11.9	

Scope of Platform Service	3,460	3,458	6.5	
Self Harm	469	469		
Violence			3.0	1

Content Moderation Appeals

Microsoft provides the information below in response to DSA Article 15(1)(d)

	Outlook.com
Number of complaints submitted to the internal-complaints mechanism	61.0

Government Requests

Microsoft provides the information below in response to DSA Article 15(1)(a).

Outlook received 23 government requests in 2024. None of these requests were related to CSAM content.